

MACRO EVENT SENTIMENT & MARKET REACTION ANALYZER

Project guide • what it is and what you need to do

Location:
D:\macro-sentiment-analyzer

Status: **Built & verified working**

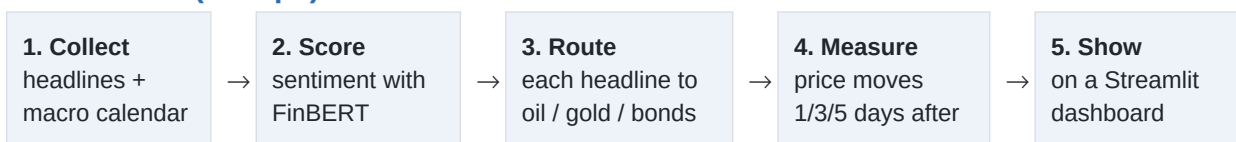
Est. time to finish & publish: ~1
hour

1 What this project is

A pipeline that reads financial-news headlines and macro events (Fed decisions, OPEC meetings, CPI releases), scores their sentiment with **FinBERT** (a finance-tuned AI language model), and then measures how **crude oil, gold, and US government-bond prices actually reacted afterwards** — all shown on an interactive dashboard.

In one line for your résumé: *“Built an NLP pipeline (FinBERT) scoring sentiment on financial headlines and ran an event study on how commodity and bond futures reacted — finding sentiment alone is largely priced in.”*

How it works (5 steps)



2 What is already done for you

- ✓ All code written & organized (sentiment model, price fetcher, event study, dashboard).
- ✓ Dependencies installed and confirmed working on your Python 3.14.
- ✓ Full pipeline run once — it scored 108 events and pulled real market prices.
- ✓ 5 automated tests pass; the dashboard boots without errors.
- ✓ FINDINGS.md filled in with your **real** results (not placeholders).
- ✓ README, LICENSE, .gitignore, sample data — the repo runs offline with no API keys.

3 What you have to do

Six short steps. The project already works — this is about personalizing it and getting it onto your GitHub.

Step	Do this	Time
1. Personalize	Open LICENSE and replace <Your Name>. In README.md replace <your-username> in the clone link.	5 min
2. Add a screenshot	Run the dashboard, take a screenshot, save it as assets/dashboard.png. The README already points to it. (I can do this for you — just ask.)	5 min
3. (Optional) run it yourself	See the commands below to re-run the pipeline and open the dashboard, so you understand it before an interview.	10 min

4. Create a GitHub repo	On github.com click New repository , name it macro-sentiment-analyzer, leave it empty (no README).	2 min
5. Push the code	Run the git commands in the grey box below (from inside the project folder).	5 min
6. Put it on your résumé	Use the résumé line + talking points on page 2. Link the repo.	10 min

Commands you'll need

Run the project (from inside D:\macro-sentiment-analyzer):

```
python -m scripts.run_pipeline # score sentiment + run event study
streamlit run app/dashboard.py # open the interactive dashboard
```

Push to GitHub (replace YOUR-USERNAME):

```
cd D:\macro-sentiment-analyzer
git init
git add .
git commit -m "Macro event sentiment & market reaction analyzer"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/macro-sentiment-analyzer.git
git push -u origin main
```

Note: the generated data in data/processed/ is intentionally excluded from GitHub (it regenerates when you run the pipeline). This **PROJECT_GUIDE.pdf** is your personal reference — delete it before pushing if you don't want it public.

4 Your key findings (memorize these)

These came out of your actual run — concrete numbers are interview gold.

Finding	The number	Why it matters
Sentiment is priced in	47.6% high-conviction 1-day hit-rate (42 events)	A coin flip — headline tone alone doesn't predict tomorrow's move.
The sign flips for rates	"Positive" news → bonds -0.46% over 3d ($t = -2.71$, $p = 0.017$)	Hawkish-Fed reads as positive <i>tone</i> but is bearish for bonds — statistically significant.
Crude reacts, but noisily	Positive-tone oil news → -1.4% next-day crude (small sample)	Commodities move more, but on too few events to be conclusive here.

Interview one-liner: "‘Positive’-toned macro headlines preceded an average -0.5% move in 10-year Treasury futures over the next three days ($t = -2.7$) — hawkish-Fed news reads as positive tone but is bearish for bonds. That's exactly why I measured reaction *windows* instead of trusting the sentiment sign."

5 Talking points for interviews

- **Nuance over hype:** "I found sentiment alone isn't tradeable — markets often price in news before headlines appear. So I measured reaction windows instead of assuming causation."
- **End-to-end skill:** shows the full loop — data → events → market reaction → dashboard.
- **Career bridge:** "I applied my existing ML skills to markets" — not "I have no finance background."
- **Method literacy:** you can name-drop *event study*, *directional hit-rate*, and *statistical significance* (t -stat / p -value) and explain each.

6 If something breaks

Symptom	Fix
"No processed data found" in the dashboard	Run <code>python -m scripts.run_pipeline</code> first.
Model download is slow the first time	Normal — FinBERT is ~440 MB, downloaded once then cached.
A price fetch fails	The code auto-falls back to ETF proxies (USO/GLD/TLT); just re-run.
Want fresh headlines	<code>python -m src.collect_headlines --source gdelt</code> (free, no key).

Research / educational project only — nothing here is investment advice. • Generated for your Macro Event Sentiment & Market Reaction Analyzer.